

Thyroid Disorders

Hypothyroidism

The vast majority of cases hypothyroidism results from an intrinsic disorder of the thyroid gland (primary hypothyroidism).

Causes of hypothyroidism

1. Causes Autoimmune
Hashimoto's thyroiditis
Spontaneous atrophic hypothyroidism
Graves' disease with TSH receptorblocking antibodies
2. Iatrogenic
Radioactive iodine ablation
Thyroidectomy
3. Drugs
Carbimazole, methimazole, propylthiouracil
Amiodarone
Lithium
4. Transient thyroiditis
Subacute (de Quervain's) thyroiditis
Post-partum thyroiditis
5. Iodine deficiency
e.g. in mountainous regions
6. Congenital
Dyshormonogenesis
Thyroid aplasia
7. Infiltrative
Amyloidosis, Riedel's thyroiditis, sarcoidosis etc.
8. Secondary hypothyroidism TSH deficiency

Incidence



Women are affected approximately six times more frequently than men

Clinical features of Hypothyroidism

Symptoms	Signs
Common	
Weight gain	Weight gain

Cold intolerance	
Fatigue, somnolence	
Dry skin	
Dry hair	
Menorrhagia	
Constipation	Hoarse voice
Hoarseness	Facial features:
Carpal tunnel syndrome	Purplish lips
Alopecia	Malar flush
Aches and pains	Periorbital oedema/myxoedema
Muscle stiffness	Loss of lateral eyebrows
Deafness	Anaemia
Depression	Carotenaemia
Infertility	Erythema ab igne (Granny's tartan)
	Bradycardia
	Hypertension
	Delayed relaxation of tendon reflexes
	Dermal myxoedema
Rare	
Psychosis (myxoedema madness)	Ileus Ascites
Galactorrhoea	Pericardial and pleural effusions
Impotence	Cerebellar ataxia
Gynaecomastia	Myotonia

Investigations

1. Serum T4 is low and TSH is elevated, usually in excess of 20 mU/L.
2. Measurements of serum T₃ are unhelpful since they do not discriminate reliably between euthyroidism and hypothyroidism.
3. The rare condition of secondary hypothyroidism is caused by failure of TSH

secretion in a patient with hypothalamic or anterior pituitary disease. This is characterised by a low Serum T4 but TSH may be low; normal or even slightly elevated.

4. Serum enzymes Raised creatine kinase, aspartate aminotransferase, lactate dehydrogenase (LDH)
5. Hypercholesterolaemia
6. Anaemia
Normochromic normocytic or macrocytic
7. Hyponatraemia
8. The electrocardiogram (ECG) classically demonstrates sinus bradycardia with low-voltage complexes and ST segment and T wave abnormalities.
9. Measurement of thyroid peroxidase antibodies is helpful

Hyperthyroidism/Thyrotoxicosis

Definition

Thyrotoxicosis describes a constellation of clinical features arising from elevated circulating levels of thyroid hormone.

Aetiology

The most common causes

1. Graves' disease
2. Multinodular goitre
3. Autonomously functioning thyroid nodules (toxic adenoma)
4. Thyroiditis is more common in parts of the world where relevant viral infections occur, such as North America.

Less common

- Drugs (e.g. amiodarone)
- Radiographic contrast media
- Iodine prophylaxis programme
- TSH-secreting pituitary adenoma
- Follicular carcinoma - metastases

Clinical features of Thyrotoxicosis

Symptoms	Signs
Common	
Weight loss despite normal or increased appetite	Weight loss
Heat intolerance	Palmar erythema
Palpitations	Sinus tachycardia
Dyspnoea	Lid retraction, lid lag
Irritability, emotional lability	
Fatigue	
Sweating	
Tremor	
Less common	
Osteoporosis (fracture, loss of height)	Goitre with bruit
Diarrhoea, steatorrhoea	Atrial fibrillation
Angina	Systolic hypertension
Ankle swelling	Increased pulse pressure
Anxiety, psychosis	Cardiac failure
Muscle weakness	Hyper-reflexia
Periodic paralysis (predominantly in Chinese)	Ill-sustained clonus
Pruritus	Proximal myopathy
Alopecia	Bulbar myopathy
Amenorrhoea/oligomenorrhoea	
Infertility, spontaneous abortion	
Loss of libido, impotence	
Excessive lacrimation	
Rare	
Vomiting	Lymphadenopathy
Apathy	Spider naevi
Anorexia	Onycholysis
Exacerbation of asthma	Pigmentation/ Gynaecomastia

Investigations

1. The first-line investigations are serum T₃, T₄ and TSH.
2. In most patients serum T₃ and T₄ are both elevated but T₄ is in the upper part of the normal range and T₃ raised (T₃ toxicosis) in about 5%.
3. Serum TSH is undetectable in primary thyrotoxicosis but values can be raised in the very rare syndrome of secondary thyrotoxicosis caused by a TSH-producing pituitary adenoma.
4. measurement of receptor antibodies (TRAb), elevated in Graves' disease
5. Isotope scanning
6. An ECG may demonstrate sinus tachycardia or atrial fibrillation.
7. Radio-iodine uptake tests measure the proportion of isotope which is trapped in the whole gland, but have been largely superseded by ^{99m}Tc technetium scintigraphy scans which also indicate trapping, are quicker to perform with a lower dose of radioactivity, and provide a higher-resolution image.
8. Serum enzymes : Raised alanine aminotransferase, γ -glutamyl transferase (GGT), and alkaline phosphatase from liver and bone
9. Raised bilirubin
10. Mild hypocalcaemia
11. Glycosuria
Associated diabetes mellitus
'Lag storage' glycosuria

MCQs

1. Which of the following cellular processes increase in response to thyroid-stimulating hormones?
 - a. No Synthesis of thyroglobulin proteins & Incorporation of iodide ions into tyrosine molecules of thyroglobulin
 - b. No Incorporation of iodide ions into tyrosine molecules of thyroglobulin
 - c. No Uptake of iodide into the follicular cells
 - d. Uptake of iodide into the follicular cells,

Synthesis of thyroglobulin proteins & Incorporation of iodide ions into tyrosine molecules of thyroglobulin

2. Which of the following is not a characteristic feature of Grave disease?
 - a) Increased total and free T₄
 - b) Suppressed plasma TSH
 - c) Exophthalmos
 - d) Decreased thyroid radioactive iodine uptake
3. Which of the following form is the active form of thyroid hormone?
 - a. T₃
 - b. T₄
 - c. rT₃
 - d. TSH
4. The subclinical hyperthyroidism is characterized by _____
 - a. Increased TSH and normal free T₃ & free T₄ levels in the blood
 - b. Increased TSH and decreased free T₃ & free T₄ levels in the blood
 - c. Decreased TSH and normal free T₃ & free T₄ levels in the blood
 - d. Decreased TSH and increased free T₃ and free T₄ levels in the blood
5. A patient presents with tachycardia and heat intolerance and is suspected of Graves' disease. Which of the following is not a characteristic feature of Grave disease?
 - a. Increased total and free T₄
 - b. Suppressed plasma TSH
 - c. Exophthalmos
 - d. Decreased thyroid radioactive iodine uptake
6. Patients with Graves' disease have high circulating levels of thyroid hormones. Graves' disease is characterized by the following, except?
 - a. Excess release of TSH from the anterior pituitary
 - b. Hyperactivation of TSH receptor in thyroid glands
 - c. Presence of TSH receptor antibodies causing receptor activation
 - d. Decreased TSH levels

7. Which of the following is not the metabolic function of thyroid hormones?

- a. Glycogenolysis
- b. Gluconeogenesis
- c. Glucose oxidation
- d. Glycogenesis

Answer

1.	d	2.	d	3.	a
4.	c	5.	d	6.	a
7.	d				